In [1]:
```
import sys
import keyword
import operator
from datetime import datetime
import os
```

keywords

In [2]:
```
print(keyword.kwlist)
```

```
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break', 'clas
s', 'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for', 'from',
'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or', 'pass',
'raise', 'return', 'try', 'while', 'with', 'yield']
```

In [3]:
```
len(keyword.kwlist)
```

Out[3]: 35

Identifiers

In [5]:
```
1var = 10  #Identifier can't start with a digit
```

```
  Cell In[5], line 1
    1var = 10  #Identifier can't start with a digit
    ^
SyntaxError: invalid decimal literal
```

In [6]:
```
val12@ = 35  #identifier can't use special symbols
```

```
  Cell In[6], line 1
    val12@ = 35  #identifier can't use special symbols
           ^
SyntaxError: invalid syntax
```

In [7]:
```
import = 125  # Keywords can't be used as identifiers
```

```
  Cell In[7], line 1
    import = 125  # Keywords can't be used as identifiers
           ^
SyntaxError: invalid syntax
```

In [9]:
```
"""
Correct way of defining an identifier
(Identifiers can be a combination of letters in lowercase (a to z) or uppercase
"""
val2 = 10
val2
```

Out[9]: 10

In [10]:
```
val_ = 99
```

Comments in python

In [12]:
```
# Single line comment
val1 = 10
val1
```

Out[12]: 10

In [13]:
```
'''
Multiple
line
comment
'''
val1 = 10
```

In [14]:
```
"""
Multiple
line
comment
"""
val1 = 10
```

statements

In [15]:
```
p = 20
q = 20
r = q
p , type(p), hex(id(q))
```

Out[15]: (20, int, '0x7ffcdf06a608')

In [16]:
```
q , type(q), hex(id(q))
```

Out[16]: (20, int, '0x7ffcdf06a608')

In [17]:
```
r , type(r), hex(id(r))
```

Out[17]: (20, int, '0x7ffcdf06a608')

In [18]:
```
p = 20
p = p + 10
p
```

Out[18]: 30

variable assignment

In [20]:
```
intvar = 10   # Integer variable
floatvar = 2.57 # Float Variable
strvar = "Python Language"  # String variable

print(intvar)
print(floatvar)
print(strvar)
```

```
10
2.57
Python Language
```

multiple assignment

In [22]:
```
intvar , floatvar , strvar = 10,2.57,"Python Language"   # Using commas to separ
print(intvar)
print(floatvar)
print(strvar)
```

```
10
2.57
Python Language
```

In [24]:
```
p1 = p2 = p3 = p4 = 44  # ALL variables pointing to same value
print(p1,p2,p3,p4)
```

```
44 44 44 44
```

Data types

- Numeric

In [27]:
```
val1 = 10  # Integer data type
print(val1)
print(type(val1))  # type of object
print(sys.getsizeof(val1))   # size of integer object in bytes
print(val1, " is Integer?" , isinstance(val1, int))  # val1 is an instance of in
```

```
10
<class 'int'>
28
10  is Integer? True
```

In [33]:
```
val2 = 92.78
print(val2)
print(type(val2))
print(sys.getsizeof(val2))
print(val2, " is Float?" , isinstance(val2, float))
```

```
92.78
<class 'float'>
24
92.78  is Float? True
```

In [31]:
```
val3 = 25 + 10j
print(val3)
print(type(val3))
print(sys.getsizeof(val2))
print(val3, " is complex?" , isinstance(val3, complex))
```

```
(25+10j)
<class 'complex'>
24
(25+10j)  is complex? True
```

In [35]:
```
sys.getsizeof(int())   # size of integer object in bytes
```

Out[35]: 28

In [37]:
```
sys.getsizeof(float())  # size of float object in bytes
```

Out[37]: 24

In [38]:
```
sys.getsizeof(complex()) # size of complex object in bytes
```

Out[38]: 32

Boolean

Boolean data type can have only two possible values true or false.

In [39]:
```
bool1 = True
```

In [40]:
```
bool2 = False
```

In [41]:
```
print(type(bool1))
```

```
<class 'bool'>
```

In [42]:
```
print(type(bool2))
```

```
<class 'bool'>
```

In [43]:
```
isinstance(bool1, bool)
```

Out[43]: True

In [44]:
```
bool(0)
```

Out[44]: False

In [45]:
```
bool(1)
```

Out[45]: True

In [46]:
```
bool(None)
```

Out[46]: False

In [47]:
```
bool(False)
```

Out[47]: False

String

In [48]:
```
str1 = "HELLO PYTHON"
print(str1)
```

```
HELLO PYTHON
```

In [49]:
```
mystr = 'Hello World' # Define string using single quotes
```

```
print(mystr)
```

```
Hello World
```

In [50]:
```
mystr = "Hello World" # Define string using double quotes
print(mystr)
```

```
Hello World
```

In [51]:
```
mystr = '''Hello   # Define string using triple quotes
           World '''
print(mystr)
```

```
Hello   # Define string using triple quotes
           World
```

In [52]:
```
mystr = ('Happy '
'Monday '
'Everyone')
print(mystr)
```

```
Happy Monday Everyone
```

In [53]:
```
mystr2 = 'Woohoo '
mystr2 = mystr2*5
mystr2
```

Out[53]:
```
'Woohoo Woohoo Woohoo Woohoo Woohoo '
```

In [54]:
```
len(mystr2) # Length of string
```

Out[54]:
```
35
```

String Indexing

In [55]:
```
str1
```

Out[55]:
```
'HELLO PYTHON'
```

In [59]:
```
str1[0]  # First character in string "str1"
```

Out[59]:
```
'H'
```

In [60]:
```
str1[len(str1)-1]  # Last character in string using len function
```

Out[60]:
```
'N'
```

In [61]:
```
str1[-1] # Last character in string
```

Out[61]:
```
'N'
```

In [62]:
```
str1[6] #Fetch 7th element of the string
```

Out[62]:
```
'P'
```

In [63]:
```
str1[5]
```

Out[63]: ' '

String Slicing

In [64]:
```
str1
```

Out[64]: 'HELLO PYTHON'

In [65]:
```
str1[0:5]
```

Out[65]: 'HELLO'

In [66]:
```
str1[6:12]
```

Out[66]: 'PYTHON'

In [67]:
```
str1[-4:]
```

Out[67]: 'THON'

In [68]:
```
str1[-6:]
```

Out[68]: 'PYTHON'

In [69]:
```
str1[:4]
```

Out[69]: 'HELL'

In [70]:
```
str1[:6]
```

Out[70]: 'HELLO '

Update & Delete String

In [71]:
```
str1
```

Out[71]: 'HELLO PYTHON'

In [73]:
```
#Strings are immutable which means elements of a string cannot be changed once t
str1[0:5] = 'HOLAA'
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[73], line 2
      1 #Strings are immutable which means elements of a string cannot be changed
once t
----> 2 str1[0:5] = 'HOLAA'

TypeError: 'str' object does not support item assignment
```

In [74]:
```
del str1 # Delete a string
print(srt1)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[74], line 2
      1 del str1 # Delete a string
----> 2 print(srt1)

NameError: name 'srt1' is not defined
```

String concatenation

In [75]:
```
# String concatenation
s1 = "Hello"
s2 = "Asif"
s3 = s1 + s2
print(s3)
```

```
HelloAsif
```